How much does a tracker-seeded ground truth favour its seed? Measuring seed bias in multi-object tracking benchmarks

Trong Nghia Le[a], Thien Bao Nguyen[a,*]

[a]3ITech, College of Technology and Design, University of Economics Ho Chi Minh City (UEH), Ho Chi Minh City, Vietnam

Abstract

Ground truth for multi-object tracking is increasingly produced by correcting a tracker's output. If the same ground truth is used for evaluation, every error the reviewer leaves in place favours the seed tracker. We measure this seed bias. A parametric reviewer model turns a seed tracker's output and the complete public ground truth of MOT17, DanceTrack and the traffic benchmark FastTrack into simulated reviewed ground truth. With four trackers on a shared detector and an assumed reference reviewer, the seed's advantage over the other trackers is 0.069, 0.160 and 0.049 IDF1, with 95% lower bounds far above a pre-specified relevance threshold of 0.01. In an exploratory analysis around this reviewer, the advantage grows almost linearly with the share of identity errors left unfixed, by 0.006–0.033 IDF1 per tenth, and an external appearance-based seed trained on MOT17 still gains 0.041 IDF1 there. Seed-correlated errors reorder the eight evaluated systems more than random errors matched in number and length and, in an exploratory analysis, also the seven systems other than the seed. Seeding from the union of several trackers helps little and favours its members; consensus seeding reduces the distortion but does not protect trackers outside the consensus; and a bracket over single-seed ground truths never contained the true score. Code and protocol are released.

Keywords: multi-object tracking, benchmark, ground truth, annotation bias, evaluation

1. Introduction

Annotating identities for multi-object tracking (MOT) frame by frame is expensive, so ground truth (GT) is increasingly produced by running a tracker, or a detector with propagation, and asking people to correct the result [1, 2, 3, 4]. The procedure is efficient, and tracker-generated labels have been shown to train trackers well at a fraction of the annotation cost [5]. When the corrected output is also used to evaluate trackers, however, every error that the reviewer leaves in place is an error that the seed tracker shares, and the seed is rewarded for it. In recent benchmarks the seeding tracker, or its association algorithm, is also among the evaluated baselines [4, 3].

The mechanism is known elsewhere. In dependency parsing, gold standards produced by correcting parser output raise the seeding parser's labelled attachment score by 5.8–6.9 points relative to human-annotated gold, and its lead over another parser by 3.6–4.1 points [6]. In machine translation, a reference post-edited from a system's output is by construction close to that system [7], and post-edits of several systems are pooled as references [8]. Quality-assessment tools for diagnostic accuracy studies check whether the index test formed part of the reference standard (incorporation bias) and whether the reference was read with knowledge of the index test [9, 10]. In tracking and video segmentation, dataset authors have noted that GT pre-generated by a model favours that model: DIVOTrack restricted labels to objects with more than 30% visibility and reviewed the whole dataset after interpolation [1], and SA-V annotated its evaluation splits without the video model in the loop [2]. Neither study measured the effect, its dependence on how careful the reviewer is, or whether such precautions remove it, and we are not aware of any study that does so for MOT.

We ask three questions. (RQ1) How large is the advantage a seed tracker gains on GT built from its own output? (RQ2) Do seed-correlated GT errors reorder trackers more than random errors matched in number and length? (RQ3) Do inexpensive annotation protocols reduce the bias? We answer them by simulation on public benchmarks whose complete GT serves as the reference, with a reviewer model whose parameters can be read as properties of an annotation process, and under a protocol fixed in version control before any experiment. Our contributions are: (i) a measurement of seed bias in MOT on three benchmarks, which at an assumed reference reviewer exceeds every difference in true IDF1 between the seed trackers on MOT17 and DanceTrack and half of them on FastTrack; (ii) the descriptive finding that, around that reviewer, the advantage depends mainly on how many identity errors are fixed (almost linearly, in an exploratory analysis), while box geometry changes it by at most 0.005 IDF1; (iii) evidence that seed-correlated errors distort the ranking more than random errors, with exploratory evidence that this extends

*Corresponding author.

to the trackers other than the seed; and (iv) an evaluation of union seeding, consensus seeding and bracketing, including negative results.

2. Related work

Ground-truth quality and rankings. Milan et al. showed that the same tracker output scores very differently against independent annotations of one sequence (MOTA 87.1 versus 68.3) [11]; their annotations differ independently of the trackers, whereas seeded GT differs in the direction of one of them. Test-set label errors are common in classification and can reorder models [12]. Re-annotating COCO masks with a model-assisted pipeline flips the ranking of several detector pairs, mostly in favour of models that predict sharper masks [13], while a re-annotation of four detection benchmarks finds missing objects to be the dominant error and reports that scores are highly sensitive to annotation quality while rankings remain largely stable [14]. These studies treat label errors as properties of the dataset rather than of an evaluated model.

Assisted annotation in tracking. Video annotation tools reduce cost by interpolating or propagating sparse human input [15, 16]; model-in-the-loop engines let a model propose whole tracks or masks for correction [17, 2]. TETA evaluates trackers under incomplete annotation [18], and SPAM trains trackers on tracker-generated labels with targeted human effort [5]. None of these analyses what happens when assisted labels are the reference for evaluation.

Pre-annotation bias. Pre-annotation speeds up annotation; dictionary pre-annotation of clinical entities showed no detectable bias [19], whereas correcting parser output anchored treebank annotation toward the parser [6], and in machine translation a reference post-edited from a system's output is by construction close to that system, which is why post-edits of several systems are pooled as references [7, 8].

3. Seeded ground truth and the reviewer model

Let $\mathrm{GT}_{\text{true}}$ be the reference GT and A a seed tracker. A reviewer with parameters $\theta = (r, p_{fp}, q, a)$ edits the output of A into simulated reviewed GT $\mathrm{GT}_A(\theta)$:

- seed boxes are matched to $\mathrm{GT}_{\text{true}}$ frame by frame (Hungarian assignment over pairs with IoU ≥ 0.5);
- each run of consecutive missed GT boxes of one object is added with probability r;
- each run of consecutive unmatched seed boxes of one seed identity is kept with probability p_{fp}; an added GT box replaces an overlapping (IoU ≥ 0.1) unmatched seed box;
- seed identities are matched one-to-one to GT identities as in IDF1; a (seed id, GT id) pair outside this matching is an identity error, and errors whose corrections interact (e.g. the two halves of a swap) form one episode, fixed with probability q regardless of its length;
- a matched box keeps the seed geometry with probability a.

With $(r, p_{fp}, q, a) = (1, 0, 1, 0)$ the reviewer reproduces $\mathrm{GT}_{\text{true}}$; with $(0, 1, 0, 1)$ it returns the seed output unchanged. The reviewer introduces no errors of its own.

For a metric M (IDF1 unless stated), the self-seed advantage of A over another tracker B is

$$\begin{aligned}\mathrm{SA}(A, B) = &[M(A; \mathrm{GT}_A) - M(B; \mathrm{GT}_A)] \\ &- [M(A; \mathrm{GT}_{\text{true}}) - M(B; \mathrm{GT}_{\text{true}})],\end{aligned} \tag{1}$$

the change in A's lead over B caused by seeding. Because the reviewer only moves GT toward the seed, the mean of SA over pairs is essentially non-negative by construction (its smallest value over all settings is -0.0001). Individual pairs are not guaranteed to be: in the deterministic setting $(r, p_{fp}, q, a) = (0, 0, 1, 1)$, SA(SORT, OC-SORT) on MOT17 is -0.023. The sign is therefore not evidence; the empirical content is the magnitude, and we test $\mathrm{SA} \geq \delta$ with a minimal-relevance threshold $\delta = 0.01$ IDF1.

4. Experimental setup

Data. We use the public GT of MOT17 (seven training sequences) [20, 21], DanceTrack (25 validation sequences) [22] and FastTrack (12 urban traffic CCTV videos) [23]. MOT17 GT was annotated from scratch [20]; DanceTrack GT was annotated with frame-to-frame box propagation that annotators refined [22], a seed shared by none of the evaluated trackers; FastTrack does not document its annotation procedure. FastTrack's eight object labels are mapped to six evaluation classes (person, bus, truck, car, bicycle, motorbike) and its remaining labels are treated as ignore regions; tracker boxes centred in an ignore region are discarded unless they match a GT box of the evaluated class. MOT17 follows TrackEval's preprocessing.

Trackers. Detections come from COCO-pretrained YOLO11x and YOLO11n [24] (input 1280 px, confidence ≥ 0.05). SORT [25], ByteTrack [26], OC-SORT [27] and BoT-SORT without camera-motion compensation [28] run with library defaults [29]. ByteTrack, OC-SORT and BoT-SORT receive every detection and apply their own thresholds; SORT, which has no low-score stage, receives detections with confidence ≥ 0.25, its activation threshold. Four trackers on two detectors give eight systems; the four YOLO11x systems are the seeds. No detector or tracker was trained on any evaluation sequence, so absolute scores are below leaderboard values. All trackers pass detector boxes through, so systems sharing a detector output identical boxes for the detections they keep.

Reviewer settings. The reference reviewer $\theta_{ref} = (0.75, 0.1, 0.5, 1)$ is an assumption, not a measurement; results are also reported, descriptively, over a grid of 192 settings ($r \in \{0, .25, .5, .75, .9, 1\}$, $p_{fp} \in \{0, .1, .25, .5\}$, $q \in \{0, .5, .9, 1\}$, $a \in \{0, 1\}$) with ten random draws each and common random numbers across settings.

Statistics. The primary metric is IDF1 [30]; secondary metrics are HOTA, AssA and DetA at localisation threshold 0.5 [31], MOTA and the number of identity switches [32]. IDF1 is pooled over all (sequence, class) units of a benchmark. Secondary metrics are computed with TrackEval [33]; IDF1 on simulated GT uses an exact reimplementation of its Identity metric, checked against TrackEval in our tests. Intervals are 95% BCa [34] over 10,000 bootstrap resamples of sequences with jackknife acceleration; MOT17 has seven sequences, so its intervals are approximate. Confirmatory statements are made only at θ_{ref} and per benchmark; a general statement requires all three benchmarks to pass. Variation between reviewer draws (standard deviation 0.001–0.005 at θ_{ref}) is small compared with variation between sequences.

Protocol and deviations. Hypotheses, the grid, the analyses and a stopping rule were fixed in version control before any experiment; the protocol, its eight dated amendments and the commit history are released with an English translation of the protocol, which was written in Vietnamese. The amendments made before any result restated RQ1 as a magnitude test, replaced the RQ3 endpoint by the paired distortion difference below, and added the YOLO11n systems of the planned cross-detector condition to the ranked set. The planned appearance-based (ReID) tracker was not run, because it needed image-level GPU inference that our setup could not provide; this deviation was recorded after the results, and all main conclusions therefore concern motion-based trackers. A planned study of real reviewers was deferred. Union seeding was first dropped on an argument that the data later contradicted and was then run, after the other results were known; its definition did not change. Analyses marked exploratory, including the external seed, were added after the results.

5. Results

5.1. RQ1: size of the self-seed advantage

Table 1 reports the advantage. On every benchmark the lower confidence bound exceeds $\delta = 0.01$ several times over, so the confirmatory hypothesis holds on all three. At θ_{ref} the advantage exceeds every pairwise IDF1 difference between the YOLO11x systems on MOT17 and DanceTrack and half of them on FastTrack. It is largest on DanceTrack, where the seeds make the most identity errors, consistent with its uniform appearance and diverse motion. It does not hinge on one tracker: without pairs involving SORT it is 0.038–0.146, and with SORT fed every detection 0.058–0.173. Changing the matching threshold of the reviewer from 0.5 to 0.4 or 0.6 changes it by at most 0.01. With car, bus and truck merged into one FastTrack class it is 0.047 [0.038, 0.060]; removing the two FastTrack classes with more than 10% class confusion leaves 0.050 [0.042, 0.060]; per class it ranges over 0.021–0.080. On secondary metrics the advantage is 0.042–0.141 in $\text{HOTA}_{0.5}$ and is concentrated in association accuracy ($\text{AssA}_{0.5}$: 0.059–0.196) rather than detection accuracy ($\text{DetA}_{0.5}$: 0.024–0.031); in MOTA it is 0.031–0.039. On its own GT a seed's identity switches fall on average to 53–56% of its true count, against 92–104% for the other trackers on the same GT.

Table 1: Self-seed advantage at θ_{ref} (IDF1; mean over the 12 ordered pairs of YOLO11x systems and ten draws), BCa 95% intervals over sequences. LOO: range of leave-one-sequence-out estimates. Cross-detector: seed (YOLO11x) against the YOLO11n systems. True-GT spread: range of the four YOLO11x systems' IDF1 on GT_{true}.

	MOT17	DanceTrack	FastTrack
Sequences	7	25	12
SA	0.069	0.160	0.049
BCa 95%	[0.060, 0.089]	[0.148, 0.170]	[0.042, 0.059]
LOO range	[0.063, 0.074]	[0.157, 0.164]	[0.045, 0.052]
Cross-detector	0.069	0.158	0.049
BCa 95%	[0.058, 0.091]	[0.144, 0.170]	[0.041, 0.058]
Without SORT pairs	0.061	0.145	0.039
True-GT spread	0.473–0.536	0.352–0.472	0.530–0.623

Reviewer parameters (descriptive). Fig. 1 varies one parameter at a time. Around θ_{ref} fixing identity errors has the largest effect: fixing every episode ($q = 1$) leaves 0.015, 0.005 and 0.018 (MOT17, DanceTrack, FastTrack), still above δ on MOT17 and FastTrack, while adding every missed object ($r = 1$) leaves 0.030–0.153. Over the whole range of r, however, missed objects matter as much as identity on MOT17 (SA spans 0.104 over r and 0.110 over q) and more on FastTrack (0.106 against 0.061). In an exploratory analysis the advantage is close to linear in q (linear interpolation between $q = 0$ and $q = 1$ is within 0.001 of the simulated value on MOT17 and FastTrack and within 0.007 on DanceTrack): each tenth of episodes left unfixed adds 0.006–0.033 IDF1, so a benchmark builder who measures q on an audited sample can read off the expected advantage. Per seed (exploratory), it is larger for seeds that make more identity errors: on DanceTrack it is 0.219 for SORT (3289 episodes) and 0.117 for BoT-SORT (1336). In all 96 settings with $q \le 0.5$ the lower bound exceeds δ on every benchmark. The retention of false positives changes the advantage by at most 0.013 over $p_{fp} \in [0, 0.5]$. Box geometry contributes little: with a shared detector it changes the advantage by less than 0.001, and against the other detector, replacing the seed's geometry by the GT geometry changes it by at most 0.005.

An external seed (exploratory). To check that the advantage is not specific to weak, motion-based seeds, we used MPNTrack [35], a graph-based tracker with appearance cues that links the public Faster R-CNN detections after Tracktor refinement, as the seed on MOT17, using

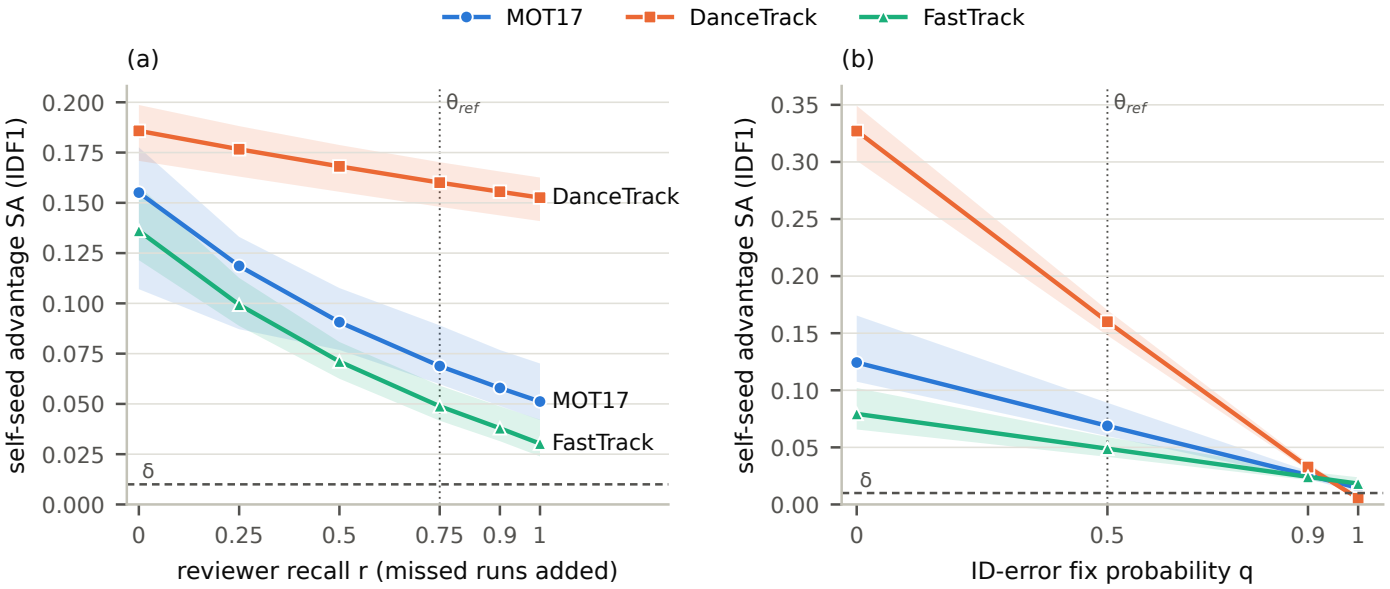


Figure 1: Self-seed advantage as one reviewer parameter varies with the others at θ_{ref} (descriptive; shaded: pointwise BCa 95%). (a) Recall of missed objects r. (b) Probability q of fixing an identity-error episode. The dashed line is $\delta = 0.01$.

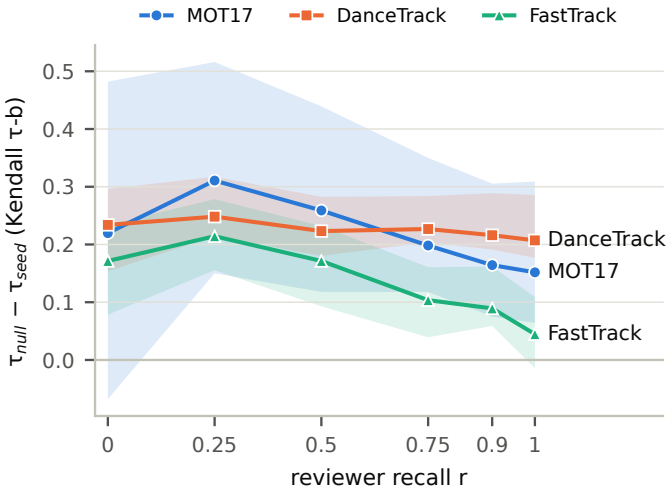


Figure 2: Ranking distortion of seed-correlated errors relative to random errors of the same number and length, as reviewer recall r varies with the other parameters at θ_{ref} and GT geometry on both sides ($a = 0$): $\tau_{null} - \tau_{seed}$ (Kendall τ-b between rankings of eight systems on simulated and true GT). $r = 0.75$ is the confirmatory point; shaded: pointwise BCa 95%.

its training-set outputs distributed with TrackEval. MPNTrack was trained on these sequences (true IDF1 0.713), so it is a low-error seed and is not ranked. Its advantage over our eight systems is 0.041 [0.024, 0.062] at θ_{ref}, 0.082 at $q = 0$ and 0.004 [−0.004, 0.015] at $q = 1$.

5.2. RQ2: ranking distortion

On seeded GT at θ_{ref}, on average 3.2 (MOT17), 3.5 (DanceTrack) and 1.0 (FastTrack) pairs of systems whose true IDF1 differs by more than δ are reversed. To separate the effect of correlation with the seed from the effect of having errors at all, each seeded GT is paired with a null GT that requests the same number and lengths of missed runs, unfixed identity errors and kept false-positive runs, placed at random on the true GT, with GT geometry on both sides. Seed-correlated errors reduce rank agreement more: $\tau_{null} - \tau_{seed}$ is 0.198 [0.118, 0.350], 0.227 [0.204, 0.284] and 0.104 [0.039, 0.161], and they cause 2.35 [1.02, 4.08], 3.17 [2.98, 3.97] and 1.05 [0.28, 2.00] more reversals (Fig. 2; the interval includes 0 only at $r = 0$ on MOT17 and $r = 1$ on FastTrack). The null could not place 12% (DanceTrack) and 14% (FastTrack) of the requested false-positive boxes, and 25–50% of the kept false-positive boxes in seeded GT carry the label of a real track, an error the null does not reproduce; the endpoint is therefore approximate on DanceTrack and FastTrack, and the null is somewhat milder than the seeded GT.

Beyond the seed (exploratory). The distortion is not only the seed moving up. Ranking the seven systems other than the seed, seeded GT still agrees less with the truth than null GT: $\tau_{null} - \tau_{seed}$ is 0.100 [0.014, 0.252], 0.145 [0.124, 0.212] and 0.086 [0.019, 0.164]. Systems running the seed's tracking algorithm on the other detector share part of its advantage: the seed's advantage SA over them is smaller than over the other YOLO11n systems in 10 of 12 seed–benchmark cases. Even with the seed excluded, the biases of two other trackers on the same seeded GT differ by up to 0.046 IDF1.

5.3. RQ3: mitigation

Union seeding. The de-duplicated union of two or three trackers' outputs contains more boxes than its first member (1.06–1.45 times for the four-tracker union), because trackers sharing a detector keep different subsets of the

Table 2: Protocols at θ_{ref}. D: mean absolute error of pairwise IDF1 differences among the four YOLO11x systems. ΔD: paired difference to single-seed GT, BCa 95%. Cost: boxes the reviewer verifies / draws, relative to the number of GT boxes. Union: range over the ten 2- and 3-member unions; members vs others: union members' advantage over the YOLO11x systems outside the union; members vs YOLO11n: their advantage over the four YOLO11n systems. Held-out: members' advantage over the tracker left out of a 3-member consensus (range over the four choices).

		MOT17	DanceTrack	FastTrack
Single seed	D	0.042	0.091	0.031
	cost	0.56 / 0.38	0.88 / 0.13	0.58 / 0.36
Union (10)	ΔD	−0.013 to 0.003	−0.024 to 0.001	−0.015 to 0.013
	members vs others	0.008 to 0.053	0.027 to 0.106	0.008 to 0.056
	members vs YOLO11n	0.022 to 0.046	0.052 to 0.100	0.017 to 0.033
	cost	0.52–0.76 / 0.28–0.40	0.88–1.02 / 0.07–0.12	0.55–0.78 / 0.25–0.37
Consensus (4)	D	0.018	0.032	0.014
	ΔD	−0.024 [−0.035, −0.017]	−0.059 [−0.066, −0.051]	−0.018 [−0.024, −0.014]
	cost	0.75 / 0.40	1.01 / 0.12	0.77 / 0.38
	held-out	−0.009 to 0.027	−0.036 to 0.087	−0.002 to 0.020
Bracket	coverage	0.00	0.00	0.00

detections. Seeding with it reduces D less than consensus seeding: across the ten unions ΔD ranges over −0.013 to 0.003 (MOT17), −0.024 to 0.001 (DanceTrack) and −0.015 to 0.013 (FastTrack), with the upper bound below 0 for 5, 6 and 7 of them; one union (OC-SORT and SORT on FastTrack) increases D (0.013 [0.006, 0.020]). The union's members keep an advantage over the YOLO11x trackers left out of it, 0.008–0.106 IDF1, with the lower bound above 0 in 29 of 30 cases. (Union seeding was run after the other results; Section 4.)

Consensus seeding seeds only boxes supported by a majority of the participating trackers (3 of 4, or 2 of 3), with mean geometry, and links each to the nearest later consensus box within 30 frames that at least two trackers, and a strict plurality, continue with the same identity; the remaining boxes are shown as anonymous candidates. It is the tracking analogue of multi-system references in machine translation [8]. It reduces D on all three benchmarks (Table 2), at the price of 14–35% more boxes to verify. Its members also keep an advantage over the four YOLO11n systems: 0.016 [0.005, 0.022], 0.028 [0.021, 0.035] and 0.012 [0.008, 0.018]. It does not protect trackers outside the consensus reliably: the members' advantage over the tracker left out depends on which tracker it is, from 0.086 IDF1 (DanceTrack, ByteTrack left out) to −0.035 (DanceTrack, SORT left out; ranges in Table 2), and on DanceTrack leaving out BoT-SORT or ByteTrack removes most of the reduction in D.

Bracketing, i.e. reporting the range of a tracker's scores over GTs seeded by different trackers, never contained the true score: every single-seed GT raised every tracker's IDF1, by at least 0.017, so the bracket lies entirely above the true score.

6. Discussion

Seed bias is large enough to change conclusions, and it reaches beyond the seed. For benchmarks built with tracker assistance this suggests a short checklist: report the seed; exclude the seed and close variants of it from comparisons on its GT, and do not use seeded GT to rank closely matched trackers, since the other trackers' biases differ by up to 0.046 IDF1 (exploratory); spend review effort on identity continuity; and measure the identity fix rate on an audited sample to estimate the advantage from the near-linear relation found in the exploratory analysis. Consensus seeding reduces the distortion among the trackers that take part, but it does not reliably protect a tracker left out of it (Table 2).

Limitations. The reviewer is simulated and θ_{ref} is an assumption; the grid shows how the conclusions depend on it, and correction logs from tracker-seeded annotation [3], paired with an independent GT, would calibrate r, p_{fp} and q. The fix probability does not depend on episode length; a reviewer who preferentially fixes long episodes lies between the $q = 0.5$ and $q = 1$ curves. The seeds are tracking-by-detection trackers. Propagation-based assistance such as SAM 2 [2] keeps the identity of each prompted object, so its errors map mainly to geometry, which matters little here, to lost objects, which matter less than identity near θ_{ref} but as much over the whole range of r, and to jumps onto similar objects, which act like identity episodes. Our main trackers are motion-based and share detectors; the external seed is a single exploratory check. The reference GTs are not free of assisted annotation; any seeding they contain is shared by all evaluated systems.

7. Conclusion

Under an assumed reference reviewer, ground truth built by correcting a tracker's output favours that tracker by 0.069 (MOT17), 0.160 (DanceTrack) and 0.049 (FastTrack) IDF1, mainly through identity errors left uncorrected, and seed-correlated errors reorder the evaluated systems more than random errors. Benchmarks built with tracker assistance should report the seed, prioritise iden-

tity review, and avoid comparing the seed and its variants on their own ground truth.

Data and code availability

All datasets are public. Code, the protocol with its amendments, and scripts reproducing every number are released at [repository URL to be added]. No video or ground truth is redistributed.

Declaration of competing interest

The authors declare no competing interests.

References

[1] S. Hao, P. Liu, Y. Zhan, K. Jin, Z. Liu, M. Song, J.-N. Hwang, G. Wang, DIVOTrack: A novel dataset and baseline method for cross-view multi-object tracking in DIVerse open scenes, International Journal of Computer Vision (2023). doi:10.1007/s11263-023-01922-7.

[2] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, C. Feichtenhofer, SAM 2: Segment anything in images and videos, arXiv preprint (2024). arXiv:2408.00714.

[3] S.-I. Nae, R. Moldoveanu, A. S. Ghita, A. M. Florea, IndoorCrowd: A multi-scene dataset for human detection, segmentation, and tracking with an automated annotation pipeline, arXiv preprint (2026). arXiv:2604.02032.

[4] X. Fang, S. Hannuna, N. Campbell, E. Simpson, 8-calves image dataset, arXiv preprint (2025). arXiv:2503.13777.

[5] O. Cetintas, T. Meinhardt, G. Brasó, L. Leal-Taixé, SPAMming labels: Efficient annotations for the trackers of tomorrow, in: European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science, 2024. doi:10.1007/978-3-031-73254-6_22.

[6] Y. Berzak, Y. Huang, A. Barbu, A. Korhonen, B. Katz, Anchoring and agreement in syntactic annotations, in: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, 2016. doi:10.18653/v1/D16-1239.

[7] M. Snover, B. Dorr, R. Schwartz, L. Micciulla, J. Makhoul, A study of translation edit rate with targeted human annotation, in: Proceedings of the 7th Conference of the Association for Machine Translation in the Americas, 2006, pp. 223–231. URL https://aclanthology.org/2006.amta-papers.25/

[8] L. Bentivogli, M. Cettolo, M. Federico, C. Federmann, Machine translation human evaluation: an investigation of evaluation based on post-editing and its relation with direct assessment, in: Proceedings of the 15th International Conference on Spoken Language Translation, 2018, pp. 62–69. URL https://aclanthology.org/2018.iwslt-1.9/

[9] P. Whiting, A. W. S. Rutjes, J. B. Reitsma, P. M. M. Bossuyt, J. Kleijnen, The development of QUADAS: a tool for the quality assessment of studies of diagnostic accuracy included in systematic reviews, BMC Medical Research Methodology 3 (2003) 25. doi:10.1186/1471-2288-3-25.

[10] P. F. Whiting, A. W. S. Rutjes, M. E. Westwood, S. Mallett, et al., QUADAS-2: A revised tool for the quality assessment of diagnostic accuracy studies, Annals of Internal Medicine 155 (8) (2011) 529–536. doi:10.7326/0003-4819-155-8-201110180-00009.

[11] A. Milan, K. Schindler, S. Roth, Challenges of ground truth evaluation of multi-target tracking, in: IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2013, pp. 735–742. doi:10.1109/CVPRW.2013.111.

[12] C. G. Northcutt, A. Athalye, J. Mueller, Pervasive label errors in test sets destabilize machine learning benchmarks, in: Advances in Neural Information Processing Systems (Datasets and Benchmarks Track), 2021. arXiv:2103.14749.

[13] S. Singh, A. Yadav, J. Jain, H. Shi, J. Johnson, K. Desai, Benchmarking object detectors with COCO: A new path forward, in: European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science, 2024. doi:10.1007/978-3-031-72784-9_16.

[14] S. Penquitt, J. Klees, A. van Betteray, P. Jashnieh, P. Stehr, M. Rottmann, L. Schmarje, Object detection benchmarks are incomplete: The role of label errors and annotation uncertainty, arXiv preprint (2026). arXiv:2609.21822.

[15] C. Vondrick, D. Patterson, D. Ramanan, Efficiently scaling up crowdsourced video annotation, International Journal of Computer Vision 101 (2013) 184–204. doi:10.1007/s11263-012-0564-1.

[16] S. Manen, M. Gygli, D. Dai, L. Van Gool, PathTrack: Fast trajectory annotation with path supervision, in: IEEE International Conference on Computer Vision (ICCV), 2017, pp. 290–299. doi:10.1109/ICCV.2017.40.

[17] P. Voigtlaender, M. Krause, A. Osep, J. Luiten, B. B. G. Sekar, A. Geiger, B. Leibe, MOTS: Multi-object tracking and segmentation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 7934–7943. doi:10.1109/CVPR.2019.00813.

[18] S. Li, M. Danelljan, H. Ding, T. E. Huang, F. Yu, Tracking every thing in the wild, in: European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science, 2022. doi:10.1007/978-3-031-20047-2_29.

[19] T. Lingren, L. Deleger, K. Molnar, H. Zhai, J. Meinzen-Derr, M. Kaiser, L. Stoutenborough, Q. Li, I. Solti, Evaluating the impact of pre-annotation on annotation speed and potential bias: natural language processing gold standard development for clinical named entity recognition in clinical trial announcements, Journal of the American Medical Informatics Association 21 (3) (2014) 406–413. doi:10.1136/amiajnl-2013-001837.

[20] A. Milan, L. Leal-Taixe, I. Reid, S. Roth, K. Schindler, MOT16: A benchmark for multi-object tracking, arXiv preprint (2016). arXiv:1603.00831.

[21] P. Dendorfer, A. Ošep, A. Milan, K. Schindler, D. Cremers, I. Reid, S. Roth, L. Leal-Taixé, MOTChallenge: A benchmark for single-camera multiple target tracking, International Journal of Computer Vision 129 (2021) 845–881. doi:10.1007/s11263-020-01393-0.

[22] P. Sun, J. Cao, Y. Jiang, Z. Yuan, S. Bai, K. Kitani, P. Luo, DanceTrack: Multi-object tracking in uniform appearance and diverse motion, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. doi:10.1109/CVPR52688.2022.02032.

[23] H. Hashempoor, Y. D. Hwang, FastTracker: Real-time and accurate visual tracking, arXiv preprint; benchmark at https://huggingface.co/datasets/Hamidreza-Hashemp/FastTracker-Benchmark (2025). arXiv:2508.14370.

[24] G. Jocher, J. Qiu, Ultralytics YOLO11, https://github.com/ultralytics/ultralytics, version 8.4.92 (2024).

[25] A. Bewley, Z. Ge, L. Ott, F. Ramos, B. Upcroft, Simple online and realtime tracking, in: IEEE International Conference on Image Processing (ICIP), 2016. doi:10.1109/ICIP.2016.7533003.

[26] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, X. Wang, ByteTrack: Multi-object tracking by associating every detection box, in: European Conference on Computer Vision (ECCV), Lecture Notes in Computer Science, 2022. doi:10.1007/978-3-031-20047-2_1.

[27] J. Cao, J. Pang, X. Weng, R. Khirodkar, K. Kitani, Observation-centric SORT: Rethinking SORT for robust multi-object tracking, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. doi:10.1109/CVPR52729.2023.00934.

[28] N. Aharon, R. Orfaig, B.-Z. Bobrovsky, BoT-SORT: Robust associations multi-pedestrian tracking, arXiv preprint (2022).

arXiv:2206.14651.

[29] Roboflow, trackers: a unified library of multi-object tracking algorithms, https://github.com/roboflow/trackers, version 2.6.1 (2025).

[30] E. Ristani, F. Solera, R. S. Zou, R. Cucchiara, C. Tomasi, Performance measures and a data set for multi-target, multi-camera tracking, in: European Conference on Computer Vision Workshops, Lecture Notes in Computer Science, 2016. doi:10.1007/978-3-319-48881-3_2.

[31] J. Luiten, A. Osep, P. Dendorfer, P. Torr, A. Geiger, L. Leal-Taixe, B. Leibe, HOTA: A higher order metric for evaluating multi-object tracking, International Journal of Computer Vision 129 (2021). doi:10.1007/s11263-020-01375-2.

[32] K. Bernardin, R. Stiefelhagen, Evaluating multiple object tracking performance: The CLEAR MOT metrics, EURASIP Journal on Image and Video Processing 2008 (2008) 1–10. doi:10.1155/2008/246309.

[33] J. Luiten, A. Hoffhues, TrackEval, https://github.com/JonathonLuiten/TrackEval, commit 12c8791 (2020).

[34] B. Efron, Better bootstrap confidence intervals, Journal of the American Statistical Association 82 (397) (1987) 171–185. doi:10.1080/01621459.1987.10478410.

[35] G. Brasó, L. Leal-Taixé, Learning a neural solver for multiple object tracking, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 6246–6256. doi:10.1109/CVPR42600.2020.00628.